

Introduction to New-Keynesian Business Cycle Modeling

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- We analyze a model that is representative of the **new-Keynesian** class of business cycle models.
- The model is *Keynesian* because it assumes that nominal prices are not flexible; they adjust gradually in response to an exogenous shock to the economy.
- The model is *new* because it is built upon a solid microeconomic foundation.
- New-Keynesian models are useful for explaining the equilibrium relationships between inflation, output, and interest rates.

The Euler Equation

- Suppose that a *representative household* lives for two periods.
- The household receives utility from consuming goods in each period.
- The *lifetime utility* to the household from consuming C_0 in the first period and C_1 in the second is denoted by $U(C_0, C_1)$ and is written as:

$$U(C_0, C_1) = u(C_0) + \beta u(C_1), \quad (1)$$

where $u(\cdot)$ is the period utility function with $u'(\cdot) > 0$, $u''(\cdot) < 0$ and $\beta \leq 1$.

Figure 1: **The household's flow of utility** in a single period as a function of consumption.

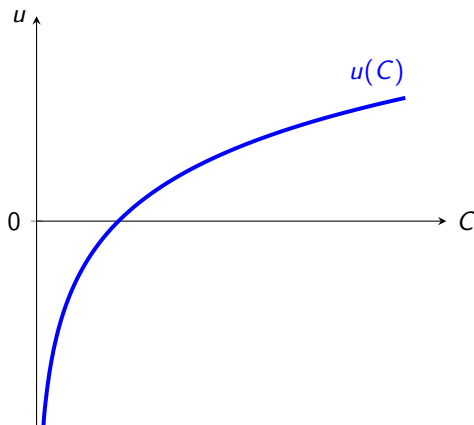
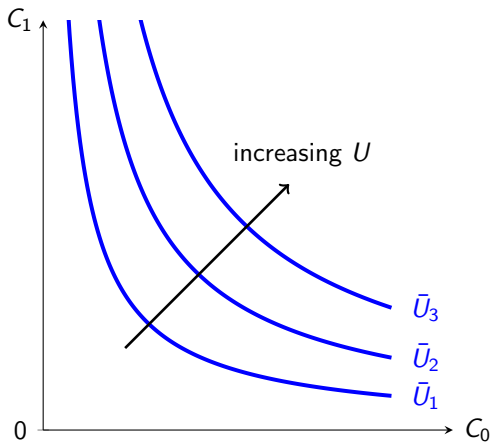


Figure 2: **The household's indifference curves** over combinations of consumption in periods 0 and 1.



The Euler Equation

- Household has no initial wealth
- Receives an income *endowment* of Y_0 in period 0 and Y_1 in period 1.
- In period 0, household chooses an amount of Y_0 to save S_1 given a real interest rate r_0 .

The Euler Equation

- The household's budget constraints for periods 0 and 1 are:

$$C_0 = Y_0 - S_1, \quad (2)$$

$$C_1 = Y_1 + (1 + r_0)S_1. \quad (3)$$

- Combine the two period budget constraints to obtain the intertemporal budget constraint:

$$C_1 = Y_1 + (1 + r_0)Y_0 - (1 + r_0)C_0. \quad (4)$$

The Euler Equation

- The household's optimization problem is:

$$\max_{C_0, C_1, S_1} u(C_0) + \beta u(C_1) \quad (5)$$

$$\text{s.t.} \quad C_0 = Y_0 - S_1 \quad (6)$$

$$C_1 = Y_1 + (1 + r_0)S_1. \quad (7)$$

- Equivalent to:

$$\max_{S_1} u(Y_0 - S_1) + \beta u[Y_1 + (1 + r_0)S_1] \quad (8)$$

The Euler Equation

- Setting derivative of the objective function with respect to S_1 equal to zero:

$$-u'(Y_0 - S_1) + (1 + r_0)\beta u'[Y_1 + (1 + r_0)S_1] = 0 \quad (9)$$

where $u'(\cdot)$ denotes the period marginal utility of consumption.

- Equation (9) is the *first-order condition* for the optimal choice of S_1 .
- We can rewrite the first-order condition for S_1 in terms of consumption:

$$\boxed{u'(C_0) = \beta(1 + r_0)u'(C_1)} \quad (10)$$

The Euler Equation

- Given Y_0 , Y_1 , and r_0 , the Euler equation (12) determines whether the household will borrow or save.
- If Y_0 is relatively large or if r_0 is relatively large, then the household will want to save.

Figure 3: **A Household that saves.** The period 0 endowment is large relative to the period 1 endowment

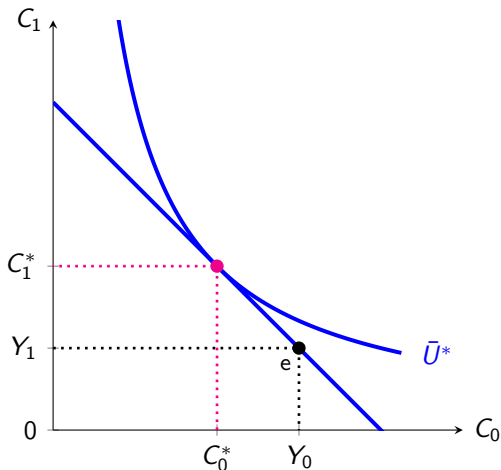
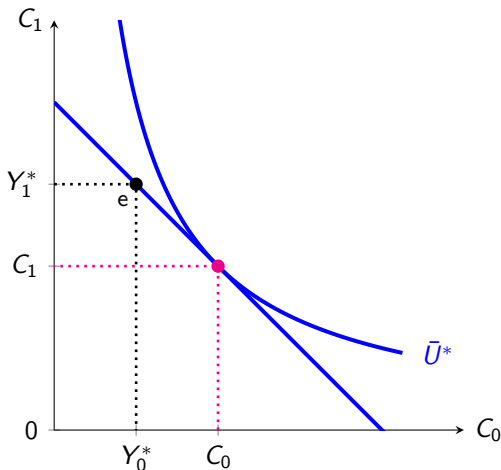


Figure 4: **A Household that borrows.** The period 0 endowment is small relative to the period 1 endowment



The Euler Equation

- We will assume that the period utility function is the natural logarithm of consumption:

$$u(C) = \log C. \quad (11)$$

- With this utility function, the household's Euler equation is expressed as:

$$\boxed{\frac{1}{C_0} = \frac{\beta(1 + r_0)}{C_1}} \quad (12)$$

The Demand for Goods

- Assume no investment, government purchases, or net exports, so $Y = C$ in every period:

$$\frac{1}{Y_0} = \beta \frac{1}{Y_1} (1 + r_0) \quad (13)$$

- Next, take the log of each side to obtain a linear representation of the Euler equation:

$$-\log Y_0 = \log \beta - \log Y_1 + \log (1 + r_0) \quad (14)$$

The Demand for Goods

- Rearranging and defining $\bar{r} \equiv -\log \beta$ as the natural rate of interest and $y \equiv \log Y$, obtain:

$$y_0 = y_1 - (r_0 - \bar{r}). \quad (15)$$

- Next:
 - The previous expression will hold for any two adjacent time periods t and $t + 1$
 - Incorporate uncertainty: replace y_1 with its expectation conditional on period 0 information $E_0 y_1$
 - Append an exogenous demand shock
- Finally:

$$\boxed{y_t = E_t y_{t+1} - (r_t - \bar{r}) + g_t} \quad (16)$$

- The complete set of equilibrium conditions:

$$y_t = E_t y_{t+1} - (r_t - \bar{r}) + g_t \quad (17)$$

$$\dot{i}_t = r_t + E_t \pi_{t+1} \quad (18)$$

$$\dot{i}_t = \bar{r} + \pi^T + \phi_\pi (\pi_t - \pi^T) + \phi_y (y_t - \bar{y}) + v_t \quad (19)$$

$$\pi_t - \pi^T = \beta \left(E_t \pi_{t+1} - \pi^T \right) + \kappa (y_t - \bar{y}) + u_t, \quad (20)$$

- The equations are: dynamic IS equation, Fisher equation, monetary policy rule, new-Keynesian Phillips curve or dynamic AS equation.